

Julian Kiverstein ~ Active Inference Insights 011 ~ Affordances, Enaction, Phenomenology

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hello everyone and welcome back to the show today I am joined by Julian Kiverstein senior researcher in the department of Psychiatry at Amsterdam medical research Professor Kiverstein's work is fascinating and deeply important as he has developed phenomenologically informed answers to a number of questions in cognitive science including time perception conceptual thinking empathy Free Will Consciousness and the self in doing so he has become a leading intellectual in the world of 4E cognitive science as he puts it cognitive scientists once shared a vision of Mind as a kind of logic engine carrying out rule-based operations on symbolic representations today this view is rapidly being replaced by an alternative vision of brain body and world as dynamically interacting equal Partners sharing the load in our day-to-day problem solving Behavior Professor Kiverstein's work has also been highly impactful on my own thinking in these domains so honestly I am really really excited to speak to you today uh Professor Kiverstein Julian thank you so much for joining me it's it's as I said it really is a privilege thanks so much for inviting me looking forward to our discussion very much as am I and I think uh you know I read I've read some of your papers multiple times and and I adore them I think some of them are absolute gems in in the four sort of lineage thank you and you sort of proposed this notion of ecological inactive cognition um now for ecological inactive cognition has a very rich history and it's worthwhile unpicking what some of these terms mean because I can imagine people are going to get confused because they're rather Niche terms at least they can sound uh as so so it'd be good to get maybe just some definitions off the ground perhaps we could start with autonomy versus heteronomy yeah uh what does it mean to be a kind of autonomous being yeah so that is an excellent starting point uh well autonomy is like a very rich concept with a long history um but the way that it's used in the context of philosophy of cognitive science is by drawing on ideas about biological autonomy so autonomy is like a form of self-organization that a living system has when it is both self-producing so that's basic to the idea of self-organization that the organization is something

that emerges and so the structure the form that the living system has is something that's emergent but also that um that that autonomous systems are able to uh keep themselves going um so maintain their organization and and that requires them to to be self- individuating in a particular way so distinguishing themselves from their environment um so I think those are the the two key features that we need to have in view the self- distinction uh self- production or self- individuation as I call it sure I've been uh I've been sort of perusing Evan Thompson's 2007 mind in life which is just cracking but also super long so I'm working my way through it slowly and here's a bit where he says well autonomy it's it's sometimes quite tricky to say when something is explicitly autonomous rather than heteronomous I think that's the right way of saying it um do you subscribe to that notion that this is kind of a spectrum or there these can we cast these sharp dividing lines yeah so there's a there's a technical discussion about whether autopoesis is necessary and sufficient for autonomy I think maybe that's what you're thinking about so autopoesis is this idea that the organization is self-produced and and there's some questions about well is that enough on its own to to get autonomy right and and Evan argues it isn't uh so you need and so does AO dep Paulo you need this extra condition of um adaptivity as they call it Y in my work with Michael kof we uh we connect that idea of adaptivity to to active inference yes so we discussed this in in the context of an issue that's come up quite a few times on your your podcast about uh to say about Pebbles or yes swinging pendulums that seem to behave as if they are minimizing free energy um but uh it still seems like uh well if free energy principle is so General as to apply to the behavior of those kinds of physical systems how is it going to be useful to use that uh the mathematics of the free energy principle to to understand uh life and M so perhaps before we we get there it'd be worthwhile just so people are aware what does someone like Evan Thompson mean by adaptivity yeah sorry so adaptivity is uh is this ability of living systems to um to regulate their encounters with the environment it requires a sensitivity to uh tendencies that can be sensed in the environment where those Tendencies could be um threatening to the precarious organization of the of the living being of the agent so adaptivity is like moving towards dynamical tendencies that are going to sustain the organization of the system and away from tendencies that are are threatening are going to Le to uh disintegration or collapse and this sounds to me like there's got piece this implies some necessary coupling with the environment whereby to use active inference terms you are there's some conditional dependencies via Mark of blanket let's say I mean that's interesting because uh as I'm as you know this vising contests with some of the earlier formulations of autonomy um so verel for example in 79 writes the environmental

elements intervening between the effector and the sensory service of uh surface of the organism is irrelevant because the nervous system can be redefined as a network of neuronal interactions in terms of the interactions of its component neurons regardless of intervening elements so that brings in this notion of operational or organizational closure perhaps it's worth explaining to the audience why at least uh on first glance that might seem like it uh engenders some notion of internalism yeah so certainly the the way that farella defines autonomy in the uh in that 1979 quote you just gave us um would entail a very strong form of internalism where the the environment is just providing perturbations to the autonomous organization of the living system I don't think in the end that is what verella intended but uh there there are places where he certainly sounds like uh it's it's a view where the the world is kind of or the meaning that we find in the world is something that's projected by the living organism onto the environment can be read as a kind of solipsism or idealism if uh without without some careful handling sure machana and verela had this metaphor of the the submarine where yeah you're you know you're underwater and and all that you have in terms of access to the external world is through the Periscope and this is sounds very similar to some of the things that we hear in active inference quite a lot where there are hidden causes out in the world and we have to infer from our sensory States what it is that are causing those sensory States because we have no direct or immediate access uh to those hidden causes yeah you know I think the quote that you just read really fits with that view of uh of life and or me um but I think that uh at the same time um we can think of the Markoff blanket as coupling us to the environment this was something that came up in your discussion with inis it separates us from our surroundings if we think of the Markoff blanket around the organism we can of course talk about where to place the Markoff blanket that's one of the things that has also come up many times in your discussions um but focusing on the the Markoff blanket arounds the the biologically autonomous system um we can think of that as separating or enforcing some kind of separation between the organism and the environment but we can also think of it as coupling the organism to the environment via the active states of the organism yes when I was going in and saying well what's the difference between the the pebble the the pendulums that are kind of oscillating together and uh and an autonomous system an agent I think a key difference is that in the case of the agent there is this adaptivity or what Michael and I call adaptive active inference uh so there's mere active inference which is what you see in Pebbles or in um the uh pendulums but there's also um what we call adaptive active inference where um that requires that the organism is able to select actions select policies that are going to um

minimize expected free energy or expected prediction errors yes yes indeed I mean again for me this fits very nicely into the kind of dissection that people make of the variational free energy equation between the so-called K Divergence um with your recognition density and your posterior so what's the difference between what you think is the posterior and what the true posterior is that's just perceptual inference but the the key thing here at least for humans is this capacity to uh maximize model evidence so this is the probability of why given your model and that seems to me to be you know well I think this is this is strongly put within the act of inference community that that that's reserved for animate active beings I those animate active beings are have what we're calling autonomy here so right that's a characterization of the organization of those systems that that goes beyond just the mere operational closure where the opponents are reciprocally dependent on each other for their production that's what operational closure means yeah and what autonomy adds to that is this idea of um of self-evidencing as we would put it in active inference terms would probably disagree with me describing it in that way but the project here is to try to bring together active inference the free energy principle with these ideas about inaction from um the inactive Community sure yeah I think a lot of these problems come very well once we make this distinction between systems that minimize variational free energy and those are abundant in the physical world systems that do this adaptively because that because of their autonomy because of their agency yes yeah yeah I see I think there are a lot of these a lot of the dichotomies or debates in active inference really come down to this fundamental issue I mean it in many ways it depends on what your lens of analysis is you have some people who focus heavily on the internal states you have some people focus on what's called the particle states so that includes the blanket states and then you have some people like a Maxwell Ram who says well really the currency is about the mark of across the mark of blanket the internal and external and that you can almost cast as you know we talk about nested Markov blankets well that itself is a Markov blanket which leads me to question you know in the classical literature about autois and operational closure people talk about the cells production of its cell membrane which relies on the metabolic activity of the cell but the metabolic activity of the cell can only happen if there is a coherent cell membrane that separates it from its environment but that again gives us this very internalist view is it is it misspeaking to say that why don't we just say that operational closure can expand beyond the skin and we could just be operationally closed in some sense with our Eco Niche is that reasonable or is that just stretching the meaning of these terms well let's back up to what you said about perceptual

inference and the K Divergence and think about how to understand that part of the free energy mathematics so in the work that I did with Yael Browner we tried to understand the difference between the actual posterior and your inferred posterior in terms of the Disattunement as we describe it between the internal Dynamics and the external dynamics of the environment so the actual posterior there is describing the external Dynamics and your uh your inferred posterior is corresponding to the internal Dynamic so then we can unpack what K Divergence is within this larger agent environment system in terms of this mismatch between the internal and external Dynamics so I think it's a consequence of that and KL sometimes says this explicitly when he's writing about Niche construction that not only is the agent a model of its environment but it applies the other way around as well that the environment is a model of agents that inhabit it so when we're thinking about um uh the environment side of things obviously we can't focus on an individual agent because that would then lead to some kind of solipsism that actually you talked about in your discussion with Carl but uh so we shouldn't think about the environment uh as predicting agents just by focusing on single agents we should instead think of group groups communities practices and so on uh as where the or how the niche is constructed and then thinking about the niche that's constructed through our uh through cultural evolution in part through practices you could start to see how there's a symmetry so that the environment could have a kind of Dynamics and external Dynamics which begins to predict what individual agents are doing yeah yes very nice yeah I this makes me think of a number of things that come into my head one is there's this thing called for some reason I've been getting kind of into the maths um to my you know I I shouldn't be but I am but in part because I recently spoke to Lance and there was a lot of mass preparation that I needed to do for that and I stumbled across something called an Fisher information uh metric are you familiar what a Fisher information metric is no cuz my m much like NE but you'll really like this the fish information metric basically it's very simple it tracks how your K Divergence is doing over time so I think this fits in very nicely what you're talking about with Attunement because you talk about you and Mark and Eric talk about prediction error uh sorry error Dynamics error Dynamics exactly yeah but that that implies over some o you know some dionic over ESS and this fish information uh metric basically tracks that in mathematical terms which is how much you know how great or how much bigger or how much smaller is your KO Divergence uh becoming over time so I thought you would enjoy that um wonderful let's go there eventually yes yeah we will yeah because that's going to connect with some other things we wanted to talk about later on about uh about

play and intrinsic motivation foraging quite so the other thing that um yeah so the other thing that came into my mind is a proposal that I made to John vvi this this one again this is an this is sort of Live question it was a philosophical pondering but I quite like it which is that our intuitions tell us that we self-evidence given um are the observations we make of the environment so and that includes inter receptive observations extra receptive and so on my proposal is that certain aspects of the environment also self-evidence given the observations that they make of us and the example that I give of that is the rock is the climbing wall so my idea here is that sure the climbing wall provides the necessary affordances for the agent of self-evidence as a rock climber but the rock climber provides the necessary affordances for the climbing wall to self-evidence as a climbing wall now this might be somewhat it might sound high futing and actually be quite dull but I think in that it it it touches upon what you were saying about the environment in being shaped uh through Niche Construction as you say can begin to manifest prediction dynamics of the actual agents within it MH yeah I mean the key thing about Niche construction is that we're we're actively modifying changing our environment in ways contribute to to prediction era free energy mization and we put these footholds in place in the climbing wall and these finger holds and in many ways that permit like that permits a wall that in its own Evolution would never be a climbing wall to become a climbing wall um yeah yeah yeah I I really like this notion well it is a you know it can be seen as uh as predicting the trajectory of the agent movements up the climbing wall right right um yes I guess the issue that a lot of people have here is structured in a way that that does um predict well that that does some of the work that otherwise would have to be done by the brain so this is like a classic way of thinking about the extended mind you asked me about the the four e and how to understand the four e and we started with autonomy but we could move to Extended mind here yeah sure and that uh that one of the ways in which extended mind is classically understood is by thinking about the offloading of computation that would otherwise need to be done by the the brain onto structures in the environment that then simplify the kinds of inferences computations that the brain has to perform so your climbing wall example I think is an excellent example of this kind of um offloading of computation onto the structure in the environment yes I so extended I've been thinking about the extended mind hypothesis uh quite a bit I I plan on doing some writing over it and I think the big debate here is where we cast what yakob har would call the evidential boundary so his notion is is that in in a very internalist way what you might say is that the mind is doing the prediction and everything else is being predicted and that's why you cast your evidential boundary because it can't be the case

that the body is both predicted predicting and predicted and then you have someone like Andy Clark who goes well um you know the parity principle if my notebook is doing as good a job as my hippocampus and that act you know that's part of my cognition and and in terms of active inference we could say well that becomes part of my predictive schema do you take a particular do you adopt a kind of fine line or a bright line as to where you cast your evidential boundary well this goes back to the the verall quote of gr actually in an interesting way because what what yakob is saying is that we should put the the Markoff blanket around the dorsal horn and that's that's the that's the candidate that he gives for where the the Markoff blanket might be relative to which self- evidencing then takes place uh so that would then imply that that anything that I'm describing as offloading is um an example of uh self- evidencing where the structure of the environment is causally very important but it isn't actually part of what's doing the self- evidencing so the the way to think about that is in terms of this notion of embedding rather than uh extending the mind or extending cognition nice uh so so yakob could well agree that um that the environment here is causally very important and the way that the environment is structured but all of the inference is now going to be done uh uh through processes that that are mediated by exchanges between the brain and the body at the dorsal horn and you know well there's many things to say about that uh so um what one thing I want to point to is a paper in Synthesia and also the the book that I wrote with Michael kof where we argue um that uh there there's reasons to think that the um self- evidencing system is an agent again and that that agent can incorporate into its body uh resources that are external to the body so that they become part of uh what's doing the the self- evidencing or what's doing the inference and so um if you think going back to where we started that the the free energy minimizing system is a an agent an agent that has this property of biological autonomy rather than heteronomy and I didn't answer your question of what heteronomy is so come back to that in a minute um but if you think that the free energy minimizing system is biologically autonomous then that system can begin to have uh an organization that incorporates uh resources that are external to the organism in the environment and those resources can include things like the structure of the climbing wall or the white lines that are painted onto the road that you use to simplify your your um your driving behavior you just follow the white lines and that gives you a very simple heuristic that you can use for for minimizing free energy yes one paper that's really informed my thoughts on this is Andy Clark's how to knit your own marov blanket it's just like a wonderful uh it's a wonderful kind of deflation of all the tensions in this area because you can just say well the markco blanket is

is an abstraction in the sense that it's always embedded within another Mark of blanket which is always embedded within another Mark of blanket and where you choose to cast that Mark of blanket as we've discussed many times on this podcast can be often just an instrumental tool and and but more importantly than that is if we do take it seriously that there is a mark of blanket it doesn't need to be fixed so we can have a process ontology rather than a substance ontology whereby when I'm in my car for me to self-evidence it's I incorporate the car's Dynamic so that me and the car it's it's sort of recursive my sort of A and B becomes a new a and we can scale this way you know all the way up to social or Global Dynamics but I think the important let's not get but I think the important point is no one's no one's asking anyone to find the one markof blanket no so there's going to be a nesting of markof blankets that's absolutely right yeah yeah always and and once you start to to see how Markoff blankets are nested then uh you can start to think about well what's the relationship between the Markoff blankets and that's where we've argued you can get the an account of how to formalize autonomy using the Markoff blanket concept because you can start to think about how the the sensory and active states that are forming the markof blanket enter into these kind of reciprocal relations of dependence that the organization of the system then depends on so you can start to get a formalization of operational closure out of the relationship between the the different Markoff blankets in a system that has a a nested markof blanket organization so that's one thing you can do also do what we've been talking about here of thinking about how some of the Markoff blankets in this complex system are going to be located outside of the boundaries of the of the organism so the sensory and active states that that are forming the markof blanket could start to uh be outside of the uh the sensorium they could so think about a spider in its web that's the example that Michael and I use as a very intuitive example of this that the the spider can start to use the movements in the web as its sensory and active states to catch prey for instance right so then the web and the spider are uh forming a single system there's a kind of extended phenotype we could talk about that forms between the The Spider and the web and also his notebook can also be described in this way so the sensory and active states that form uh the boundary of the extended cognitive system can could now be uh could now involve the notebook and the writing down of things in the notebook the looking up of information in The Notebook B and so on yes yeah see the thing is I like that my my only sort of naive reticence is because we're conscious and this is such a sticking point but I think where some of the cortico centrism comes from is the kind of presumption say this before but you know that Michael and I argue for extended Consciousness so we maybe we get there a bit

later on there's one more thing I want to say about the the extended mind and how um active inference is relevant to the extended mind discussions um and this relates to the idea of recruitment of external resources and how recruitment works so for a long time Andy Clark's argued that the the brain is is a kind of um lazy brain that tries to simplify its computations as much as possible for evolutionary reasons so you could relate it to the principle of least action probably this idea of efficient computation um and sometimes to efficiently compute it makes sense to offload that computation onto the environment so this is classic kind of situated cognition idea that goes back to the 1990s um maybe even further back than that um so how does that uh how do you then weigh up what's going to be the most efficient way to solve a problem do you solve the problem just by relying on your onboard biological Hardware or do you offload the problem solving onto the world and recruit external resources so this is called the recruitment problem because it's a problem about how you soft assemble a cognitive system on the fly in such a way as to uh as efficiently as possible Compute a solution to a problem and uh what Andy Clark's been arguing in recent work is that the the recruitment problem is solved by uh your uh your um imperative to minimize prediction error over time to minimize expected prediction error or expected free energy so we can give a kind of account of how it is that cognitive systems whose boundaries extend beyond organism and come to recruit certain external resources to solve problems in terms of uh free energy minimization or in terms of expected free energy minimization that can become a policy that's inferred that the most efficient way to solve a problem is by recruiting external resources that simplify the computations that would otherwise have to be done inside of the hge that's really really nice I'm writing that down because I really like that uh cool that's that's awesome okay yeah there's a paper that we can link you to in the australasian Journal of philosophy that Andy published um I think last year um there's a there's a reply to yakob Hoy as well called busting out yes yeah yeah yeah that also makes these these same sorts of arguments that and in that paper Andy ties it to uh precision and to precision as kind of sculpting the effective connectivity of the brain in a context sensitive way yeah so he has the idea that when you're Computing Precision sometimes those computations will lead you to to increase the gain on um on well I'm going to put it in terms of the the sensory and active states that are external to the body uh that relate to the structures in the environment so then the as you put it before the Markoff blanket is not fixed it it kind of extends to include those resources in the external environment although that said there's a there's a way of modeling sort of um a basian graph for example or a pom DP scheme whereby that which you deploy Precision waiting over is not necessarily

incorporated into your own markof blanket so there's a really nice paper by mitz and colleagues 2019 where they talk about selective attention and how selective attention needs to be governed by the capacity for cognizers to to pay or or to downregulate task irrelevant objects let's say but that doesn't mean that the task irrelevant objects are automatically Incorporated within the self markof blanket it's just that they have gain control or there's gain control over them um no but I was thinking a different case where it's not so much the task irrelevant objects that you're now dis attending to but the task relevant objects that you're attending to and how the Precision mechanism can be what explains how those task relevant objects get to become uh a part of an extended cognitive system uh so it's the opposite of the case that you were describing where they were describing how task relevant objects are not selected yeah I'm thinking about the case where a task relevant object that is external to the body like the book or the spiders web are uh now attended to because they are uh precisely because they're task relevant they allow you to to solve a problem in a much more efficient way than you would if you just relied on your brain to clarify um I I was s of introducing it as the kind of downrating of task irrelevant um objects but it's it precisions are zero some game so up of the gain so oh I see okay I guess it's all relative on the task relevant stuff sure sure I think I won't misspeak people can see it themselves but the outcome is the same um in the sense that it's all relative so precisions are zero some game so if it goes down here it's going to go up here saying is you don't want to include the task stuff yeah yeah yeah yeah so that's exactly their point yeah yeah you do want to include the task relevant stuff that's what the soft assembly of the cognitive system is it's the it's the selecting what's relevant to your problem solving in this context I guess my question here is does it make a difference whether we downregulate task irrelevant objects because in doing so right if we just think of a landscape where everything is equal in pressing these down you get relative gain on those which have not been suppressed versus actively inflating the Precision of task relevant objects do we not get the same outcome just in relev uh relative terms so I think what's super interesting here is that Precision turns out to be doing what John VY cause relevance realization yeah so uh the model that you've just been talking about is so which prediction errors should be treated as Salient or uh reliable for so that they get to influence future processing um and so you you are focusing on the the error signals which are downway so not treated as Salient because their task irrelevant I've been talking about the error signals which where you turn up the gain because they are Salient sure they are relevant so this is part of the the problem of how uh as VY calls it how do we solve the in a particular context the problem

of what's relevant and what's irrelevant yeah the frame problem yeah yeah yeah yes no you you might indeed you might well be right the mid tole 2019 with Frist and frame it that way again it's been a while since I read it but it it converges on the same point which is that the human organism the the cognizer has the capacity even before it even glances over even before it does the iser card to have this disambiguation which I just think is stunning and I think you can scale that up towards the fact that we you know in in the classical um bias competition Theory they just desimo and Duncan basically just postulate that go being uh go representations inflate that which uh you know inflate the stimuli that correlate with the goal so if I'm looking for my Ginger friend you know I have this template of Ginger and everything Ginger start everything orang or Ginger starts in increasing in game control what inference gives you is a really really nice I think it gives you a really nice way of modeling that although I don't think it's been done yet which relies on sort of deep parametric modeling and about um mental action yielding Precision waiting over sensory expectations I might have a mental action okay I want to uh have high Precision over anything that's Ginger and then in turn I'm gonna act so as to confirm that I think the other thing just for the audiences and something that has really become quite clear to me is there's difference between precision and precision waiting and this is something that I think needs to be articulated a little bit more in the active inference literature in so far as Precision is just the inverse entropy of a probab probability distribution or you can just say it's standard deviation but Precision waiting is actually the deployment of uh well it could be considered gain control but gain control is not just based on the Precision of a signal because when I'm so if the Precision of the so they people talk about attention being the optim ization of What's called the a matrix which is the likelihood distribution so what are the CH given uh given a state what's the likelihood of this observation but just having a really strong onetoone mapping between those two doesn't mean you're going to pay attention just to that because if I'm looking for my Ginger friend there might be loads of things out there which have very precise mappings to their causes but I'm ignoring them so they might be very precise observations but they don't matter um so that's just to say that I think people I think at least it needs to become more articulated within the literature exactly what this difference is although the as I said the Deep parametric modeling of it hasn't been done yet as far as I'm aware so there's plenty of work to be done there but yeah John this is right up John Street okay what you said there about it if you're interested in finding your friend with ginger hair that's the thing that matters to you so that's Vision I think that's really in in an important point because it it makes the connection between attention and uh and your goals y

yeah and it also makes the link between attention and what you care about so what matters to you and yeah as an agent so finding your friend matters to you that's something that you care about yeah yeah yeah friend so there's a link here between precision and and emotion which I think um in the work I've been doing with Mark Miller we would argue leads to an embodied view so going back to the ease now we could think about the E of embodiment um we can also think about the E of extendedness some more here but thinking about the E of embodiment the reason that we should think of precision as embodied is because what you care about is sculpting uh modulating what is given Precision in a particular context so in your example what's given Precision is finding your friend that's uh and and that's given Precision because that matters to you because that's important to you because right you care about your friend because we're because we're autonomous because we act according to our own norms and values of which which we've made that's right that's part of it h so um people talk about autonomy as then entailing uh making or meaning making because uh based on your autonomy uh there are then Norms meanings uh values that become intrinsic to your biological organization uh so in this case the value of your friend you're caring about your friend can be thought of as something that falls out of the uh free energy minimizing Machinery um because that Machinery is the Machinery of uh an autonomous being so values meaning and so on become intrinsic to uh the sustaining the maintaining of the organization of that being sure I'm GNA I'm going to spit more on idea and I'm curious about what you think about it so people take this idea of the relationship between attention and values and inflate it quite strongly and I don't mean that in a disparaging way because I think some of the work is very powerful but for example Le someone like a Jordan Peterson or Jonathan Pou or even John vvi might say well what is sacred the term sacred is that which you pay most attention to or that which cap or that which governs your attentional schema so for example you could see we have a hierarchy of values so why do I want to earn money because I want to go on holiday why do I want to go on holiday because I want to be happy why do I want to be happy and then Aristotle would say well that's just that's it you got it do you think that we can do you think that's a viable way of recapturing the idea of the Sacred if that means anything to you in terms of computational attentional mechanisms wow that's a big question I know so uh how do we get to to wisdom and the sacred and so on from uh from this kind of computational Machinery uh I think there there's an interesting story to be told about that um and I think we can get there uh but it requires a number of different steps sure along the way to connect attention as it's uh modeled using Precision to uh attentional schemes as you just described

them that could be shaped by uh religious beliefs for instance in the beliefs in sacred and so on um and no I think we can go there because um this free energy principle once you think of Markoff blankets as nested yeah can be applied to very complex um groups communities by thinking about how um the agents that make up that group are now part of collective systems uh where it makes sense to think of those Collective systems as also minimizing expected free energy so one of the routes I would go here would be to think about cultural ecosystems and think about what you just called the the sacred uh and the the sorts of attentional schemas that are involved there in terms of these these larger Collective um free energy minimizing systems yes yeah exactly move I would make here and going back to the individual MH um and how an individual may have an experience of the Sacred how to make sense of that well uh you're getting into very deep uh aspects of Consciousness when you think about uh subjective experiences of the Sacred and and very deep Notions of meaning making um and your friend example that that got us into this was was an example of a meaningful well a source of meaning in people's lives which is comes from the the con the social connectedness between people um that that's a source of meaning U but obviously the sacred is also a profound source of meaning for people and but it's a very thick concept so how to make sense of the kind of meaning that people get from the sacred uh the meaning in life that that gives people uh using these these ideas of predictive processing would require a much more complex story than how the value that we place in our uh our connectedness to one another um that could be traced back to something that's more I think to do with primary emotions the or basic emotions to go to the the emotion literature something that we see in animals as well and that probably is tied to neurotransmitters but the sacred um the kind of meaning that people find in the sacred to account for the way that that gives meaning to people's lives in terms of predictive processing I think can be done but it requires a much uh more complex story yes I think I think some of that story will come one from your notion of nested affordances but not that these nested affordances are separate from one another there's a very important idea in predictive coding uh literature that the higher levels of the predictive coding hierarchy contextualized and constrain the activity at the lower levels so for example I have a and the higher levels track longer term regularities so they're slower they're more entrenched and they constrain and contextualize what happens uh with the tracking of slower faster fluctuations to give an example uh I might I've written this actually in this new paper that I've written so my beliefs that I want to go to France in June is based off a higher order the belief that summer in France is quite a nice experience and that is going to constrain every action that I do so it's going to constrain the

way that I go on Skyscanner and book my flight but then the other important thing is that those lower levels also interact with the higher levels so again I give this example if it rains every day in June in fact I'm probably gonna take out an umbrella on July first despite the fact that I had this high order belief that it's always sunny in France in the summer so I think this this um prediction this trickling prediction down and the cascading or the uh the movement up of prediction error is maybe where this answer is going to lie why as you say why we get this notion of meaning in Life or the sacred or this particular phenomenological Essence I think is up for grabs I don't think anyone knows as of yet but I think it is going to be done in this kind of deep in the modeling sense this deep ometric modeling so something AK I think what I like about what you just said was that um when you think about the experience of the Sacred it seems that there's a whole Rich story to be told there about what the sacred is exactly so you're looking at a narrative of some sort and there's a good argument for thinking at least in terms of the self model yeah what you find at the the higher layers of the self model are going to be narrative type structures that connect your self in the past to who you are in the present where you're going in the future so where are you going on holiday in your example but uh so it's the there's a narrative structure to uh both to the experiences of the Sacred I would argue and and to uh what you're uh doing when you're planning your summer holiday right and so how would do we think about a narrative well you could think about that just within the individual in terms of for instance the the mental time travel idea the mode Network that is involved both in Imagining the future and in remembering the past so you could think of the process of constructing a narrative as just something that happens within the individual or you can think no when it comes to very complex experiences like the experience of the Sacred the construction of the narrative there is something that involves a much larger system uh a collective system where the uh beliefs about the sacred that are used to make inferences for instance are ones that uh we need to uh explain in terms of some dynamics of a larger Collective system uh so what's at the top I think is what I'm asking you here you have the idea that that the top layers of the generative model are constraining um the decision making the SEL inference of of policies at the lower levels and that's that's absolutely right but what's at the top can be thought of using the idea of Andreas rüper and uh and Chris FR in terms of what they call uh top top interactions so they can be thought of in terms of interactions that take place within cultural systems or within a cultural ecosystem uh so to make sense of this kind of um uh experience in terms of uh the inferential processes that minimize free energy we need to look at a a much larger Collective system here not just a a system inside of the individual and

that's another way of thinking about extended mind where we're thinking of Mind here as distributed over uh a cultural ecosystem not as something that's in side of individuals yes that's wonderful that's wonderful yeah so I've been thinking a lot about these self models but I think you're absolutely right the sacred Taps into self- evidencing of almost a world model um a a world model that is not not for one but for what you call what vicenin and you go and and you call a way of life form of EX form of life form of life way of life I love as well a form of life form of life I I was literally thinking I'm not going to mess that one up okay a form of life in the sense that a form of life is it's connected with a way of life and certainly the sacred is connected with the way of life sure but yeah go IDE form of life where we Define that as um as a a regular pattern of activity that can be found within a a group or a community right so perhaps it's a notion that scales but you can apply it at a collective that's the the key idea that's relevant here a species as a form of life or as members of that form of life so individuals as forms of life and when we think about autonomy again biological autonomy that's that's a way of thinking about what form of a form of life is at the scale of an individual organism so it's a scaling notion yeah yeah yeah like the Markoff blanket is a scaling notion because that it can be applied to very big things like economies and cultural ecosystems and it can be applied to very small things like cells or um or molecules particles yeah this is this is super interesting again yeah exactly scale is really the relevant thing here the question is what what what is at the highest level of of the form of life's generative model and there are answers that some people are posited I mean Consciousness um there's there's this idea at least um I don't think they would have expanded this to like a panus view but it certainly lends I would say some it leans quite heavily on pan psychism you know Maxwell RAM and Ma alaras in this notion that Consciousness is this the uppermost Mark of blanket in brain well this is not too dissimilar to Someone Like Jordan Peterson who says that you know the voice of God is actually the voice of Consciousness and and that's the guiding spirit so again we're postulating but I think there's a really interesting framework set up there for anyone who wants to get into that which is what is the fundamental thing that's constraining us and contextualizing us and I don't I just don't think it's survival I don't think it can just be biological imperatives I think maybe it's just me being romantic but I I sense that it transcends that um yeah so there are two things I want to say on that before you move on so sure sure please the first is that going back to the ROP sto and fifth paper about the top top interactions what they were interested in that paper is that the idea that what you have at the top in top down control they were thinking about executive functions for instance is um something that

in the prefrontal cortex for instance that might be the source of will or willpower or agency and they were arguing that actually it makes sense to think that um that the executive functions could be uh unfolding not within individuals but within larger groups or within diads or within social systems so so the the role of uh executive functioning or top- down control would then be done in interaction with other people rather than something that unfolds within individuals uh so that that's a way of thinking about what's at the top in terms of topdown control or executive functioning where we think think that the executive functioning is now the control processes are uh in the sense of the thing that might be doing willpower is something that is social and in involves social interactions rather than something that happens inside of the head oh that so nice oh that's yeah I like that okay that's y that's really good that's one thing I want to say about that and then the Consciousness being at the top in top down control well again we have this idea of willpower as involving conscious intention making or something like that and and then there's all of the arguments about whether the the wheel is epip phenomenal or is it just an illusion like Daniel Wagner's work and um and and then people that want to say no it will the will is tied to Consciousness and or people who want to say no all intentional action is just generated unconsciously and there isn't really any work for the will to do anymore based on like liit type arguments and so yeah there is a connection between Consciousness and and this idea of executive functioning uh not it's a bit different from what Maxwell and Ma are doing I think that there is a connection there maybe it ties more closely with what Adam saffron tries to do where he tries to bring this like herous idea together with Free Will and that's part of his work yes I mean I mean verel said what that we're composed of micro identities um I think there's always an open question about how seriously you take this notion of uh self-construction once you start getting to the very the very top or the very bottom level of the generative hierarchy so what a minimal self model might look like um but that's that's that's a different question maybe we'll come to self modeling so we I wanted to finish the thought on the will on Consciousness sorry because I didn't quite get to the end of it but yeah the thought that the the path I was on was that um was going to take us back to narrative again actually so um so if we think about the narrative layers of the self model I think there's there's a good good reason to think that they they might be where you need to look to understand uh will for instance will power yeah in that um so Elizabeth paser in her work on um on intention makes this distinction between present directed and future directed intentions which are like what we call plans in in active inference the future directed intentions and those know you can think of as coming from your narrative about who you are those future directed intentions like

your example of going on holiday to France that's a future directed intention that then sculpts the decisions that you make when you go onto your uh your flight booking oh sorry can we just pause a minute sorry about that no not all uh so we're talking about present directness and future dirting this exactly um and how the narrative might be what uh connects together your present intentions with your future intentions and your your past experiences I think your example of holidaying in France Illustrated that very well so you have the past experiences of France being beautiful in the summer and that then feeds into your uh your plans for the future leading you to form particular intentions in the present absolutely so so the narrative we can think of as uh as at the higher level doing the work of controlling planning in a way that you might think of gets you intentional action so actions which where you could uh think that the intention is the cause of your actions which is what we're looking for when we think about Free Will that's that's excellent yeah yeah yeah absolutely I completely agree um yeah there be some wonderful ideas here um so just want to point people in direction of uh yakob lonowski work with Carl they sort of point to uh the depth of temporal modeling Carl has done this in his solo work as well to be really what underlies the construction or the generation of what they call an epistemic agent model and this is something that I spoke about in in the paper which is that an epistemic agent model can use uh past experience past knowledge to inform allostasis counterfactual reasoning and all these things we associate with planning so that's but that all comes from this really wonderful work that car done with with people like um Jakob lowski okay that's yeah that's that's excellent wonderful okay we we we're talking exactly the same language which is which is awesome now I want to go on to something maybe a little different and uh I'm gonna make a I'm gonna say a quote firsty uh just to frame the conversation so Lawrence Shapiro who you'll be very you know you'll be very aware of his work he says I challenge anyone to explain how I managed recently to travel from Madison to Cologne purely in terms of my sensitive responding to Natural signs now he says this in response in a review of hotto and Mayan's books so uh the 2013 and the 2017 book now I just finished the 2013 book uh and I I and I've been delving a little bit into the sorts of critiques so evans's critiques uh Lawrence's critiques but again I'm not an expert um in my conversation with Chris I actually had to go back and sort of clarify but Harter and may make this basic point that basil or basic cognition is contentless representation free um but you do have these higher order forms of cognition which are scaffolded of social interactions symbolic systems developmental schema uh which are basically about public language use now you if I'm speak correctly have a skepticism of this distinction between higher and lower forms of cognition

and you in your paper on affordances 2014 say we suggest in says you say we find this commonly made distinction between lower and higher forms of cognition unhelpful we suggest instead that the relevant distinction is better marked by differences in level of ability or expertise in doing things with the affordances of the environment so my first question is do you think that cognition is representation free uh and if so to what degree is it all of it is it just the lower order the basic cognition what what were your thoughts in that domain yeah so representations we're going to go into representations so I've been getting really into it and I know it's a very hot area and people get angry at each other so we're not no one's going to get angry um and feel we don't have to go into it no I'm not going to get angry certainly not uh it doesn't make me angry but the the issue comes up when we think about the generative model and and and what exactly uh BAS and priors are so there's this purely statistical understanding of what basy and priors are and and what they turn out to be in the generative model and that I think we can unpack without bringing in anything like representations or content um this is a controversial position so most people would think as soon as you've got a internal model of the world then that internal model has to be a a representation uh because they think well it meets all of the conditions for represent ation like for instance there's a structural isomorphism between the model and the environment where that isomorphism can be exploited uh so made use of by Downstream processes to infer to make decisions and so on uh so it looks like the internal model the generative model meets all of the conditions for being a a representation a full-fledged representation and I think it does doesn't I think it's just a mathematical construct the generative model uh that uh can be used to model autonomy um and um we can think of the the agent not as having a model yeah so as um instantiating and a model of the internal uh sorry a model of the external Dynamics in its internal Dynamics yeah that's what I've been saying yeah but um rather as um just a mathematical construct that describes the Dynamics the self-organizing Dynamics of the agent in its coupling with the environment so we can give a contentless uh content free representation free account of what the generative model is that's part of what I want to say here MH now you ask me uh does that then commit me to something like hat and Mayan's distinction between basic and uh cognition which is non-representational doesn't involve representations and uh and non-basic cognition which is where which is content involving or representation involving yeah and to answer that question we need to know what representations are because they clearly exist um so it's just the issue is whether it makes sense to think of the brain as having content bearing states are so where exactly do representations come in and there I think we need to look at Classic kinds of

examples of what representations are in human life so we find representations when we uh use pictures and images and diagrams and those are uh are coming from cultural practices like uh the use of diagrams to do science or the use of figures in right writing articles or uh the use of um uh pictures when we're doing art uh that can be traced back to making paintings on Cave walls um so these are um representation based practices um but they come from culture they don't come from uh what the brain is doing uh so that's one example of what a representation is that it's an image or a picture a second example of what a representation might be is a a state that carries semantic content so what is semantic content well it's um a type of meaning where uh there's a determinacy to that meaning such that you can specify truth conditions conditions of satisfaction that the state that's carrying that uh Mantic content can be evaluated for so the state can be correct or incorrect it can be evaluated as correct or incorrect true or false and so on and the problem with the the kinds of states that we find within a generative model is that they don't have the kind of determinant contents that map onto uh semantic meanings so when it comes to um making the translation from Shannon information mhm to semantic information where semantic information has this determinacy um this specificity to it there's a there's a mismatch uh we have to explain how to get from what is mere co-variation or correlation which is what you're dealing with when you're thinking about Shannon information yeah information that's specific enough that you can start to say that the the organism could be wrong in how it's taking the world to be or wrong in uh the the way that it states are states that are have meaning for the organism so that it's that mismatch that makes me think no it's just not right to to think of the generative model as uh in terms of representation we can just make sense of it as a mathematical object that is tracking statistical patterns captured by the sorts of variables parameters that we um that we use to write down the the generative model long answer but hopefully it gets some of the issues on the table for us no it's wonderful it frames it frames it very nicely and and that is precisely actually in that 2013 book at least what Harto and Mayan say that just to repeat what you said that you there it's It seems impossible to go from co-variance to content uh and I think they actually have a principle for it because why not um yeah yeah of course lots of philosophers don't agree with them about that so Nick Shay who's who's a good colleague of of Chris Fris and they've worked together he has a whole book where he lays out how you can get uh semantic meaning from uh adding some extra conditions to co-variation and right he thinks you need both indicators so that's what Shannon information gets you and also consumers of those states that indicate something very roughly and so once you have like a richer account he thinks that you

can give an argument for thinking well no there there is real semantic meaning here not just um but that's one thing he wants to say yeah sh also has this like more nuanced picture where there's just different kinds of content that we need to bring in and uh hat and may really focus on a very rich demanding sort of content find in language that a paradigm example of representation and what what Shay tries to do is show well there's more U there's less demanding Notions weaker Notions which still qualify as content yes okay well yes so and Chris didn't have too much time for the radical in activists um most most uh cognitive neuroscientists have no time for radical in activists and and it's their own fault because the radical in activists are so kind of polarizing in how they uh engage in this discussion right so kind of yeah exactly so I have no skill in the game so I'm not gonna get no I have no passion here but I think I think so I think Harter and Mayan obviously spend a lot of time criticizing um this notion of cognition as content or the mark of the cognitive being representation where I think there are viable critiques of their book and their work is the alter alternative is somewhat not particularly at least explicated in any great depth um they say for example the idea that organisms are so they they want to endorse the idea that organisms are informationally sensitive are ye that they exploit correspondences in their environments to adaptively guide their actions while denying that it follows that they take in store or process informational content now this notion of exploiting correspondences sounds very much like there's some there's a function there which indicates something else now that is representation right under a dreian account of what representation is now on the other hand if they say well no all we're talking about here is just the responsivity of the organism the the adaptivity the good jobs that they do by responding to stimuli well then I think this is the point that Shapiro makes or maybe it's Heaven this is this could just their teio semiotics what they call their teos semiotics is no different from behaviorism and actually it's even more radical than the behaviorist because the behaviorist all the behaviorist said was well it's just not really part of the scientific purview to talk about internal States uh because all we can observe a behavioral um you know behavioral action they didn't even they didn't necessarily deny um content bearing States they just thought it wasn't worth the examination until Mr Chomsky and so do you like is there a way out of this um conundrum that one it's we find it very difficult to get content out of covariance and there is coari but then the other hand it's very difficult to account for adaptive Behavior solely in terms of covariant so uh on the behaviorist thing and I think H and may well certainly Mayan is willing to just bite the bullet on Behavior oh really fantastic so behaviorism is often depicted as this like devilish position that nobody should hold

anymore since the cognitive Revolution but sure some caricatures and Straw Men about what behaviorism is exactly that uh I think May in at least is in the business of and also Louise Barrett and uh some of the work that she's done they're trying to uh correct for mistaken interpretations of what behaviorism actually means so that is part of the U the radical inactive project at least some people have pursuing that within radical in activism um so how to go well why is indication that serves a particular function not sufficient for uh for Content according to hat and M U um why couldn't we use that model and just have a completely contentless uh explanation of behavior well they think it's not sufficient because it doesn't get us the spe specific Take On The World world that's required to say that an organism could be mistaken and any any definition of uh content or meaning uh requires that that that the state that's the carrying the content could be mistaken it could be misrepresenting how things are in terms of its uh that State's particular Take On The World and so when we try to build in specificity and get that out of function yes that there's uh there's long-standing problems that that project has run into again it fails to get us the specificity that we require so where the debate's kind of got to and this is Nick Shay's position is that you just need to to have a more relaxed view of what specificity is and such that it depends on your your question that you're asking what kinds of questions you have in doing psychology or cognitive Neuroscience well dictate uh whether you need more or less specificity uh so you relative to your questions you can get the kind of specificity you need to talk about content we shouldn't think that there's one size fits all picture that's going to answer all of all of the uh the different questions that we might have because each question might require a different level of gr granularity or specificity sure well not to try and offer a one-sized fit all but I'm curious about uh your position so moving sorry I've been saying what's been going on elsewhere and the discussions not no no no no not at all that's my fault as well I it's all the framing which is excellent and I also wanted to backtrack on the hard turn myio thing because uh it was it was dealt with quite briefly in my conversation with Chris and there were certain things I had to go in and clarify post Haw so it's useful so people you know are aware of these clarifications um so what what do I think about all of this well I'm sympathetic to the idea that that we when we think about what representations and content is then we should this is where my phenomenological background comes in yes we need to go back to lived experience which is where the notion of intentionality was first introduced by brentano and H other phenomenologists um so intentionality has its origins in lived experience um and then when when we try to deploy this notion in doing science um then we've taken up a particular attitude towards the world the

scientific attitude um but the notion of content and meaning itself derives from uh from uh our perception and thought as conscious beings uh so there's a bigger question here about how to uh make phenomenology fit together with the uh the naturalistic world view that we get when we start to do cognitive neuroscience and um yeah this goes back to another one of the ease the ease of the inactive uh Camp uh where going back to the embodied mind of verella Thompson and rash from 1991 MH the critique that made of cognitivist cognitive science a cognitive science that thinks just cognition is information processing of some kind yeah was that it doesn't have room for Human Experience if you think of uh of cognition as information processing how do we then account for uh the what's described phenomenologically when we think of intentionality as a structure of Human Experience expence so intentionality here means the directedness of the Mind upon objects and the idea of intentionality is that you you're always working with a a kind of relation between subject and object so intentionality is supposed to describing a a mode of experience that involves a essential relationality between subjects and their world heigo went on to talk about this in terms of being in the world talk with in a beautiful episode thank you that was wonderful yeah there's loads of things that come up here I mean so first you're not going to find an enemy of yes please go ahead if you got yeah just uh know where I look for intentionality and meaning is in lived experience and then that raises the question of how to bring together how to make room for what was described in phenomenology ology when they were describing the human mind in cognitive science in cognitive Neuroscience um and verella Thompson and Ros put to put forward this idea of mutual constraints that what is described phenomenal logically uh is a constraint on the kinds of explanations that we give in cognitive science and what's described in cognitive science likewise constraints that's why this is a mutual constraint what we can make sense of phenomenologically so there's a kind of um what's described by uh verella in terms of um a what's the term he uses now Maxwell also uses this in one of his papers on computational neurop phenomenal oh yeah it's um there's a kind of generative um passage between phenomenology and cognitive science that that you get from this Mutual constraint yes yeah yeah yeah uh yes I was saying yes he says from generative models to passages exactly yeah so you're not gonna find generative but but the idea is that we ought to be able to have this kind of um passage so bridge that goes between phenomenology which was not describing anything in a scientific way but trying to go back to lived experience there ought to be like a a bridge that takes us from phenomenology to what we can now model computationally using the free energy principle that's the idea of generative passage well that's the game that I'm in um and

enjoying that's the game I'm in as well and it's fun yeah I'm enjoy involved in this within the active inference Community yes it gives a wonderful way of doing it back to um know to to the original embodied mind idea it that book The embodied mind was what I found as an undergraduate that that brought me into this field um and it's very inspiring it tries to bring together phenomenology Buddhism and cognitive science uh through this idea of inaction um I still think it's a brilliant book I would recommend it to everybody Evan wrote it when he was young and verella was U later stage in his career so some of the ideas I think are developed much better in in mind and life which you also mentioned earlier um but but just because of its kind of broad brush Strokes it makes it very readable and and so everybody can kind of get on board with the project at least sure yeah I'm very much enjoying mind and life um yeah so as you know you're not going to find an enemy of phenomenology here um I like my haiger and hle and Merlo Ponte I particularly like like draus um and draus has a draus I think is an important addition to this conversation because I hate to go back to them but well I don't hate to go back to them but Hatter and may actually mentioned draus dra's notion of skillful coping skillful action as exhibiting intentionality and bear in mind yes it's not basic cognition according to them is not contentful but it is still intentional in some sense it's targeting um but they use d to say well he doesn't there are no mental representations or um content and I build off some of this in this notion of flow states which I think are a really example uh interesting example of which which Taps into some of these ideas but the problem with this is that draf I don't think and I think this is the point that actually Evan Thompson made that draf wouldn't say that those states are contentless and I wouldn't say those states are contentless rather they're intentionally contentful but in a way that's not representational in a way that it's not like there's a representation which stands for something precise in the world to which you can attribute uh well that function you could attribute a a a truth claim rather there are these Norms that are governed by perceptual perception and action Loops um and this biological autonomy that we've been discussing and but ra but there's still this intentional content which he would call skillful coping um and I think you could consider to be this kind of Auto tocity that comes out of flow so what does what what what does the what do the phenomenologist contribute more broadly Beyond draus although he does rest his arguments a lot on Merlo Ponte um about the role of content yeah so um let's first of all go back to the the categories of the ready to hand and the present at hand that yeah that dfus really focuses on in his uh in his work uh interpreting division one of hiiga Being and Time um so when we're dealing with the the ready to hand uh um we're behaving in a way that our our actions are are drawn from us by the

solicitations of the world that's the kind of phrasing that he often and gibs and that we Eric and Eric geld and I and Y Browner uh the three of us have been very much inspired by Gus's reading of the ready to hand um so it's this responsiveness to the solicitations so solicitations are for us uh affordances which stand out from uh the surroundings as relevance or meaningful salience to what the agent is doing so the classic example that haiger uses which you know from the conversation Marlin is the the hammer and the nail I think you're a little bit worried about that example actually well just en tiing it to flow and every day yeah yeah but it's fine it will do yeah so you know but we could think about my just reaching for my coffee cup yeah I want to take a drink of coffee the the coffee cup solicits my behavior it draws my hand towards it so um so my actions are kind of drawn from me in my forming an equilibrium state with the the the environment and so in our in our work with Yer and Eric we Tred to argue that this tending towards an optimal grip can be thought of as a consequence of uh of what we were then calling variational free energy minimization and now I think since then know that that paper was written in 2016 so the field has moved on enormously now we would think about that those sorts of examples of tending towards a good grip tending towards an improved grip in terms of expected free minimization um but the idea there was that that this kind of tending towards the right sort of optimal relationship with the environment uh is a consequence of uh what we can model in terms of free minimization um now is that representational well I've been saying no it's not because the the the generative model is is not something that the organism has it's something that the organism embodies right we can think about what are the Dynamics that the organism instantiates and do we need to bring in content uh uh or representations to uh explain the states that the the internal states that the organism instantiates and that's where I think you know the idea of inactive active inference sorry inactive inference that ramstead and uh Carl and Michael introduced in their adaptive Behavior paper um Carl talks about embodied uh inference sometimes these are all ways of unpacking what how active inference might be understood where you think no the organism is a model and so it doesn't need to represent anything it can sure it can just be thought of in terms of uh states that carry Shannon information about the world where that's non semantic and so that's one thing I want to say that the translation from the ready to hand via this notion of solicitation tending towards an optimal grip those Notions can be un unpacked uh in a way that um maybe we can carry ize in terms of content content to be motor intentionality as Mel Ponte was describing it or compartment towards the ready to hand as heiger would describe it yeah if that's what you're calling content fine yes it's okay to talk about content here why

because the content here is traced back in the end to something that we can describe in lived experience sure I wouldn't call it representational and nor would I call it like semantic meaning right draus wouldn't either uh those satisfaction conditions Max does so Maxwell has this paper where it's he's trying to give an account with inis actually strangely inis is like a nonrepresentational inactive theorist but they have this paper together where they try to give a formal semantics for they try to derive a formal semantics ICS out of um basian mechanics yes and know that formal semantics if you think you can get formal semantics out of basian mechanics I think that that's really a a representationalist program that you're on there sure thing they do is kind of it goes into this instrumentalist realist thing again that you've talked about before yes so there's there's a little bit of as if stuff going on or looks like you got to read you got to read carefully you got to read through the lines yeah um so perhaps if you go instrumentalist here then you can hold on to in's kind of sure inactive take on where representations come from that they only come from practices they're not something that you find or they come from the science of making models so scientists make models those models can then be interpreted instrumentally as uh useful tools for representing Dynamics and so on so you can start to think think of where representations come in through scientific practices or without thinking that the brain is is doing uh operations over semantic bearing content bearing States sure I mean I I still find it a slight challenge that distinction because I think there's a question about well how does how does public language become exapted into uh internal language such that you can even speak of internal representations but I think that's a separate question um I think the thing to say there as well and this is something I've argued with with Eric as well that if you think that that the generative model is non-representational it doesn't thereby become representational just through a process of uh reenacting or reusing uh states that are uh content bearing like uh when we engage in inner speech and we make use of right language to think with that's very much a phenomenological experience that we have of thinking in language um but we shouldn't think uh infer from that that when we're explaining inner speech in terms of neural processes then um there's inference over semantic bearing States going on the so yes yeah I in have the issue separating those phenomenal and content but maybe that's just me there's the modeling which is uh something that can be characterized in purely mathematical uh statistical terms just in terms of Shannon information and then there's the interpretation the gloss that we give of the model as scientists where we may be able to interpret what we're modeling in terms of meaning and representation but that's something that we're doing as scientists who are making use of representations to interpret

their models and yet there is still unless you're an Illusionist about the whole thing phenomenal experience and so I find it difficult just personally to to separate content from phenomenal so yes that's what I was arguing as well and trying to connect to motor intentionality in in phenomenology in or to the ready to hand okay dealings with the ready to hand so yeah okay okay that okay I see that okay those are those are Notions of intentionality and content and meaning yes without without truth and accuracy being built in yeah and they're they're not they're descriptions of lived experience of human beings uh so uh when we're describing lived experience I agree it makes total sense to bring in the notion of intentionality as a structure of experience then if we try to read that back into what we're modeling when we use a generative model or um or into the target system that we're modeling the brain yes then there's a an extra move that's being made there that we can go from the notion of intentionality as it's used to understand lived experience to whatever we're doing as scientists when we use a a Model Model A Target system the brain so there are three things there there's lived experience there's the model and the basian beliefs that we use to uh to write down a generative model and then there's the target system the brain that we're using the generative model to uh to make inferences about in the end um and uh where do we bring in intentionality well for me only in lived experience and then we can use the model the basian beliefs basian mechanics to uh explain in the terms of cognitive neuroscientists in the terms of cognitive Neuroscience sorry uh lived experience that's the generative passage from phenomenology to cognitive Neuroscience that's made possible by basian mechanics but we shouldn't think that then we're going to read back straight from the phenomenology uh intentionality into the basian mechanics basian mechanics is just mathematics and statistics yes explains what's described in phenomenology because of the generative passage but we need to be careful not to just conflate the two and mash the two together they're not the same thing pH mechanics and intentionality no it's a very useful disambiguation I guess the the only question that well one of the only questions that comes one of the questions that comes to mind because I have plenty uh is about perception because perception is cast as resulted again it's quite it's not particularly well sketched out but resulting from basum mechanics in the sense that perceptual adequacy is just the minimization of the KO Divergence of the recognition density and the true posterior so this leads us quite nicely into affordances because you have in your 2014 paper you have I believe the section is called afford affordances for higher cognition and you talk about this example of a nettle and mint leaves which I would like to read out because I think it would be a very useful framing so you say um so you quote

Gibbs who says that the affordances of human behavior are staggering because it's in the whole domain of social significance and then you give this example you say for instance when we are out looking for mint leaves to make mint tea and a friend incorrectly reaches for a nettle we can stop him or her by making the judgment that is not a mint leaf that is a nettle in doing so we are skillfully engaging with the affordances the nettle leaf has in our form of life the leaf of for judging correctly that it is a nettle in our form of life you see for me for me I just cast that as a that just for me is just well one person's model is perceptually adequate and the others isn't and for me that's pure Bayesian mechanics about the reduction of the K Divergence but that yields that doesn't necessarily yield me an affordance in terms of public language use it just yields perception so maybe you could explain because people might hear that and go well I don't think that the seeing of the uh nettle is in and of itself encapsulated in the notion of an affordance to inform your friend that it's a nettle rather this is just perception right and the role of perception is purely just to achieve what I've been calling descriptive adequacy whereas action the maximization of model evidence is normatively gives us normative viability normative uh yeah normative explanations so it seems to me that you're kind of slotting perception into um norms and in some sense the maximization of model evidence I.E I need to be a good social actor in this situation well one one thing that's important about that example is that it's supposed to be an example of discursive uh thought so the person judges that's not a uh a nettle sorry leaf yeah that's not a a nettle that's a mint leaf or is it the other way around I can't remember no it's not a n it's not a mint leaf it's a nettle but the point is that it's a it's a demonstrative judgment that's being made a judgment about the the type of leaf that the person's just picked so there's a there's a perceptual categorization going on that might make it look like it's just like a a perceptual uh phenomena that's being described here but it was being introduced as a an example of perceptions serving as a a reason for making a judgment uh a particular kind of demonstrative judgment uh about the the type of leaf that you're looking for and whether this is that type of leaf or not uh so uh uh it was being given as an example of higher cognition because uh people think that higher cognition is reason responsive so you're in the space of reasons you're not just in the space of causes anymore to use u a distinction from Wilfred Sal um so if you can make this kind of demonstrative judgment based on your perception that's because you're you're responsive to what the reasons are here uh so there's a kind of rationality in play in this example of perception um and then the question is well can you you were pushing the idea well it's just about the uh the Bayesian mechanics this kind of using of perception to make judgment um and there I think we need to

be careful to distinguish personal level explanation which is explanation in terms of reasons from subpersonal processes which are being described by basian mechanics uh where uh the inferential processes here are supposed to be just a matter of um mechanics just a matter of basian mechanics sure um and we want to in the end connect up what we're appealing to in our personal level explanations this is the bridging idea uh with what can be described subpersonal in terms of basian mechanics but how do you go from the personal level explanation which is reason giving kind of explanations to subpersonal explanation uh there's an interface problem as it's sometimes called here how do you get your reasons to interface with causes which is what you're interested in when you're thinking about mechanics and I I think what you were doing was saying well you were missing that there's an interface uh that we need here that allows us to move from U personal level explanations to subpersonal explanations and we shouldn't lose that interface no no I agree with that uh I I've been I've said on paper that I think there is an issue how perception let's say just gets chunked out of ko divergences I that's an issue I'm not denying that I guess my question here was here so going back to the phenomenology uh discussions again uh the way I was we were thinking about this was well there's the kind of intentionality that perception has where uh that can be thought of just in terms of U what H called passive and active synthesis so there's a kind of Constitution that we can appeal to at work in perception that is uh pre- predicative comes before any categories are introduced and then there's um a kind of intentionality which is distinctive of thought and judgment where where that is uh predicative it's categorized so you're bringing in some kind of subject too distinction you're dealing with the present at hand yeah um and this example was supposed to be like an example of that moves from the ready to hand to the present at hand and because there's some sort of breakdown that happens there's a mistake that's made um that gets corrected and that pushes you to start to think and make judgments uh so you're not any longer dealing with just being drawn by well actually we want to say you are because we bring in this idea of affordances that come from language but um but what's important is that we're dealing with a with a discursive propositional predicative form of intentionality in the Judgment that's being made here not just a pre- predicative subject objectless form of intentionality sure yeah I guess I guess my part of my point was uh we have to integrate the ready to handedness form of intentionality with the present at handedness intentionality abely yeah and I I I think like maybe one reading of this would just be like oh we're just skipping over the fact that there's some uh uh concept free perception going on in the first place and I just wanted to get clarity that that's not what you're saying right there is still the perceptual act

prior to the categorization that language grants us so there's there's like I agree with you that we need uh some way of integrating the ready to hand and the present to hand that's what you just said that's absolutely right that takes us back to this question of higher and lower cognition um how we want to call into question that distinction in the end um so you could think there's still a distinction to be made here so looking at the paper by Andrew Ceran and Jakob Hoy and Giovanni Pulo yeah they have this beautiful picture where you can go from um homeostasis to allostasis to planning deep temporal models and you might think that underwrites a distinct fiction between higher cognition which is where you find planning and deep temporal models and lower cognition where it's just about homeostasis or about chemotaxis like in the Shanon colleagues model of chemotaxis so there's this like differentiation that you get where higher cognition emerges in in these deep temporal models um yes and what we want to say is well that's a kind of approach where you try to get to higher cognition vertically by building up from um lower cognition via homeostasis to higher cognition deep active inference uh parametric deep parametric active inference as L. Saned Smith calls it quite yeah yeah now we want to do something that's different from that which is to horizontalization instead of trying to build up to higher cognition vertically in the generative models through temporal depth we want to look at the Horizon uh in which we engage with the environment and think that uh spatially and temporally there can be more or less distal engagements with the environment that are made possible by us being uh by our cultural practices so this goes back to where we actually started on the question of cultural ecosystems Niche construction and how um once we have in view the idea that the environment predicts agents as much as agents predict the environment then we can start to have a picture where instead of thinking of uh temporal depth as coming from um the hierarchy in the Deep temporal generative model we can think of temporal depth as coming from the ecosystems in which we're situated um and the way that those ecosystems make possible dealing with distal uh states of Affairs that are currently absent from us that's what I think we get from language from uh being able to um engage in propositional type uh reasoning reason in that where we're using predicative forms of intentionality this is how you sort of end up swallowing up yeah to to do um perceptual judgment to do thinking about what is currently absent um dealing with the distal and so on where people normally bring in representations yeah so this is how you swallow up if you pardon the pun uh representation hungry cognition uh um whereby you dis you sort of refute offline cognition as being distinct but rather this quote more complex form of coordinating nested states of action Readiness and activities to multiple relevant affordances such a process

is complex because of the nesting of the activities and their increasing reach through time which I guess gives us well at least your answer to Shapiro's uh challenge which is how is not going to be convinced by that I mean you already said well yes natural signs don't explain how you can navigate from Madison to wherever yes I guess my question here is why can't we have both and so what I mean by that is you also give another example from Shapiro reenacting his past experiences going to his house his former house in New Jersey and you say that you know him pretending to engage in those activities of being there is in itself a Act of coordinating to multiple relevant affordances but my question is why can't we H why can't we sort of have our cake and eat it too in the sense that we can we can accept that and we can accept that when I'm trying to get my train from or my plane from Madison to Cologne uh that's me reacting to the affordance of going to cologne and what that might afford me why can't we also say that to even kick off that process I have this kind of online offline representation which frames my nested activities I want to go to Cologne or are we or or is that permitted because you want to reframe representation Hungary cognition you you say you know you give the quote possibility of a positive alternative namely whether the representation hungry cognition might be achieved by other means by for example skillfully coordinated complexly structured nested activities to multiple affordances such that episodic memory imagining can be given in these terms what I just don't see why we can't have have both we had the benefits of saying okay these are representational so they include content they include public language or private language in this case and also there a response to affordances why do we need one rather than the other um so it goes back to a number of things that we've been talking about already so one is where does intentionality originate from uh where does content meaning originate from from and so yeah uh I I've been contending that uh that the source of intentionality is lived experience um sure so the reason why I want to appeal to uh practices in the form of life uh like how language is used within communities following a vicentini model um is because that traces meaning back to uh something human to to uh to what human beings do when they are speakers of uh English so um what they are able to do then is to articulate certain thoughts but they're also able to um to revisit past experiences uh by constructing narratives about those past experiences so uh that connects us to the The Narrative topic again um and I think the way that you were describing things was to to suggest that narratives can be modeled at the higher layers of these deep temporal models that's absolutely right they can be modeled in that way um but does that mean then that that the narratives just are uh the uh processes processes of inference that take place within these deep

temporal models and that's where I I would say no we shouldn't make an identity between the narrative and the uh and what we're modeling using a model because the the modeling practice is being done by a scientist yeah and those models that the scientist is constructing are what we use then to explain some Target phenomena which is the the cognitive process of constructing narratives um but can also be understood in terms of the link the actions of uh Norm normatively regulating your linguistic behaviors so that they fit with how other speakers of your language behave um that's something which is very much a cultural communal activity um so uh yes generative models capture these processes they give us ways of explaining them in the terms of the free energy principle and cognitive Neuroscience um should we then take the phenomena that we're modeling to be nothing over and above uh what can be modeled using these generative processes so we simply identify the behaviors of constructing narratives thinking about our past experiences being a master of linguistic practices uh can we just identify those behaviors with what we're modeling when we use generative models that's where I say no because there's a lot more to um the linguistics behaviors uh there are the practices the what people do when they belong to communities now interestingly and this connects to where we started from those practices and um communal behaviors can also be modeled using uh basian mechanics where we think that the system now is is a collective system that has maybe collective intelligence and is made up of lots of interacting agents uh so we can also model that system using basian mechanics but should we then say that the collective system just is uh a free energy minimizer uh no because the S the target system here is one that can also be described by doing social science by doing Linguistics um it can be described using lots of different explanatory Frameworks and all of those planetary Frameworks can be made to um Bridge can be bridged to from basian mechanics that's super interesting but we shouldn't then try to do a reduction of all of these other explanations to basian mechanics um so what I'm resisting here is a kind of reductionism uh where everything can just be taken back to physics again sure I want to say no there are social dynamics cultural dynamics that uh where we need the tools of social science where we need to do ethnography for instance or anthropology or where we need to do phenomenology to understand these Dynamics and and those also give us very good ways of interpreting and understanding what human beings are doing as agents yes yeah I I I didn't mean I I don't think it's a reductive claim the phenomenal the relationship between the phenomenal and the physics um where I've written about it I've used words like emergent or causally Downstream from and I know these are problematic words and whether it's supervenient or not these are obviously very tricky and philosophically

um deep questions I guess my my only Point here about was if it it doesn't need to be reductive to say that representations at least in this uh episodic memory reimagining might have a role or may play a part I'm not saying it's the whole picture I'm not saying that it replaces the basian mechanics I think representations do play a role in episodic memory because there are forms of episodic memory which are are such that we can articulate and we can express linguistically what we're remembering and that's process of of putting it into words of articulating it is part of the con construction the Reconstruction of that memory okay yeah that's that's exact yeah that's exactly right so so then the qu my question really was why are we considering this a you you posit the uh the framing of offline representations as nested activities governed by the affordances that the environment grants us as a positive alternative to representation hungry cognition but it sounds like you're also permitting there to be some representational schema in place so yeah the differences that we don't think of the the representations as something that's part of the information processing but uh representations get involved but not not via the generative model because the generative model for us is just is just a mathematical model it's just pure statistics it's not bringing in and you can then give interpretations of that statistics using constructs like basian belief and inference over beliefs and inference of posterior and so on but that's all a gloss on something that is just mathematics and statistics so where do the representations come in when we're trying to understand and explain in terms of cognitive Neuroscience episodic memory they don't come in through the brain they come in through the coupling between the agent and an environment that is structured by social cultural linguistic practices so we talk about affordances in our more recent workers in language Andy Clark talks about material symbols he has very interesting work with um um trying to think of the name the guy's name Gary lupian about how U material symbols for instance can sculpt Precision expectations um but they're doing that work not uh internally but uh as material symbols that are uh in the environment that are situated in the environment so this example of language sculpting attention sculpting Precision is an example of a looping through the world where the material symbols are being taken up and included in the um in the Precision estimation process in the sculpting of precision as something that is located in the environment something where the agent is coupling with a with an environmental structure in this case the material symbols yeah that that that makes it a lot clearer for me I guess the question that going back to Shapiro's episodic memory imagining so what would that look like in this kind of worldly Loop so he so just to provide the example he imagines going back to his family parent house in in New Jersey and he's say how on Earth

could that be you know how on Earth can that be representation free but you're not saying that's representation free you're saying it's just that that representation is yielded from the ACT uh the agent environment relationship and it's not actually part of the basian mechanics okay but what does what what does that mean for this the basian mechanics because I want the basian mechanics story to apply to Extended cognitive systems as well sure sure it's just internal to the brain anymore the part of this that's getting the representations involved is the coupling with uh representational practices so being a language speaker who can uh can articulate and put into words uh a past experience like the um the poster that was hanging on your bedroom wall as a child yes Jimmy Andre being able to remember that poster is something that you can do because you're a language speaker who can uh can describe can make sense of linguistically past experiences that you had is it wrong to say that just because that that capacity is scaffolded from the my development and the culture from which I've been in which I've been raised that that hasn't become internalized is it or you say are you permitting that because you're saying it's not in the brain I mean my intuition is telling me that fine like language didn't I mean unless you know we're not going to go into in nativism and chompsky but like because it's not really that relevant but fine language comes from uh exposure to data and societal norms and so on but at a certain point I can use language in the absence of social contexts uh in ter like in the case of my em imagining Shapiro's em imagining so it seems like in that case it has been internalized to some degree it's just that it it was the the presence of a previous social Dynamic was necessary for its internalization MH yeah so that's really interesting and the the key question there is how to think about what internalization means exactly in the case of episodic imagining um so uh in keeping with the ecological inactive approach uh what we what we try to do is say well what conceptual framework do you need for making sense of uh episodic imagining in terms of uh basian mechanics so uh is it a representational conceptual framework or can we instead use the uh the tools of ecological inactive cognitive science the what we call the skilled intentionality framework to make sense of um the basian mechanics that's involved in in in that example of episodic imagining um and we would argue that uh you can make sense of What's Happen happening in that case by thinking of it not vertically I.E looking at the top layers of the generative model and then saying that's where episodic imagining happens it happens in a kind of mental time travel that is made possible by Deep active inference but instead thinking that the cognitive system here is an agent environment system so uh episodic imagining internalization of linguistic abilities is a matter of skills that you have as a as a language user and those skills are uh are then well

the kind of meaning that's involved here is accounted for in terms of responsiveness to affordances so um this all just sounds very philosophical and not interesting to scientists so I'm not sure how much your listeners are going to get out oh they're they're philosophers they they love it um I mean there's yeah to go to agent environment system and and to try to understand um horizontally in terms of the Dynamics within the agent environment system what could be going on when somebody can revisit their past experiences and and construct that kind of experience of the poster on their wall uh so horizontally what's going on there we think is that you're using your skills as a language user um so as a master of English I can then um make sense of my uh past experiences um and and and interpret them as the experiences of the this poster of Jimmy Henrik let say yeah yeah oh okay cultural stuff that's packed into that yes and that that stems from our skills as speakers of a language sure okay to cap off this profound conversation into affordances which I'm sure has taken many people into the weeds they probably us talking about it I guess the question is the overarching question therefore is so these representational states are scaffold yielded from language learning in a in a cultural setting in terms of the actual act therefore of reimagining are is the presence of those do the presence of those representations have a causal role or are they just epiphenomenal over the um the affordance exploitation ah but the representations just are affordances for us so now we try to have an ecology a cultural ecosystem that is populated by uh affordances that that grow out of our uh linguistic practices and so um so the representation external representations we think there there are no internal mental representations because um those are scientific constructs yeah I an internal mental representation that require you to be able to solve this hard problem of content that we were talking about earlier on yeah um we don't think that representations are not real there are real representations but they're just not internal mental representations they're not internal to the brain they come from our cultural systems like our linguistic practices our systems of making pictures and images and diagrams and we couple to those practices as participants in those practices and that's what allows us to engage in uh in representational style cognition by making use of these external resources in our environment okay and I guess the people's worry might be the sort of standard objection that this position means that affordances can't have any objective quality which undermines Gibson's original realism but I believe you have a response to that in terms of the fact that we also have to take in the material reality of the affordances to into account not only our sociocultural practices again the paper the 2014 paper's cracking so everyone can just go there um I mean how long have we been going for 2 hours and 5 minutes okay we'll wrap up in a bit

in a sec there's so many things I want to ask about um we didn't even really touch you can you can like figure it out in the editing right yeah yeah yeah yeah yeah it's all it's all in the editing in the end I don't edit them that much um I do want to so there's there's a paper that you have about mindness so about self-modeling and mindness um which I thought was really interesting as I've say I'm F this is slightly further down on your Google Scholar but I saw it and I thought I have to have a bit of that because I love the self modeling is entirely what I'm fixated on the moment not entirely and you say in line with the arguments of the phenomenologist I will claim that every feeling must be felt by someone it must have mindness built into it if it is to feel a particular way um some people might go well I thought that Anel Seth and Absen Airus said that mindness came from inter receptive inference and that is the sort of seat of the so-called minimal phenomenal self and if that disbands so does the notion that the experiences the feelings the affect I have of mine um maybe it's worth just separating your perspective or your position from theirs yeah so this goes back to again generative passage between phenomenology and uh and free energy principle cognitive Neuroscience because the the mindness construct is a phenomenological construct it's describing a structure of um of Consciousness whereby every conscious state is also experienced at the same time as it is an experience mindness is capturing the relationality uh that is part of the the experience whereby it's an experience of mine or yours it's sometimes this is called sense of ownership um that each experience is owned by a subject um and then when you have phenomenal like depersonalization or yeah uh derealization uh that can be thought of in terms of uh a loss of this experience of uh of mindness we can debate whether that is entirely lost or not so Anna Chica says it's not it's just Disturbed in some way so there's still this phenomenological structure that's present but the phenomena of depersonalization derealization is a disturbance of that structure of experience and I think that's right um so then what we need to try to do is to uh to take the phenomenological structure and and Bridge to how we could think about that in in terms of um active inference um and that's where the idea of interceptive inference or multisensory integration understood in terms of um integrating interceptive predictions with extracephalic that we can use to build that bridge from the phenomenology the idea of mindness as a phenomenological structure to cognitive Neuroscience okay so okay so so yeah Anna doesn't say that it uh completely disbands the uh sense of ownership I don't know if you've read George Dean's paper on well he's got two papers wonderful one on derealization the other psychics yeah self dissolution he goes a bit further and I make this point at the end of my paper that I think they're talking about two

separate parts of the D realization process and I think Anna's is just a bit more moderate whereas George's is a bit more severe um it's a good good difference that you pick up on there that's that's definitely a a uh Divergence an interesting Divergence just talking well in my from I think it's very explicable that Divergence but I think they're just talking about two different phenomenological States but George would say she would be a really good guest she yes well yeah I've spoken to her plenty of times I think yeah I think she she might well be coming on um but I guess the question here is okay so I'm from the school of people who say well mindness is emerging from the computational processes it's emerging from the inter receptive inferences from the integration the uh the multisensory integration so what that means is that if that breaks so for example George gives example of torture so and I mentioned this to Chris as well that if you're in a torture chamber um you're going to be very unlikely to be able to succeed in suppressing inter receptive prediction error and torture is a very reliable instigator of depersonalization but you but there are still experiences there are still feelings but the sense of ownership has gone but you said I will claim that every feeling must be felt by someone so who are those feelings being felt by so what's lost is the transparency in that case so um ordinarily our uh our sense of mindness the experiential self is not something that gets in the way something that comes conspicuous in our experiences rather transparent we look through it to the Wells and that notion of transparency is then modeled using uh uh dis attending sensory attenuation Thomas metzinger all that yeah yeah so uh what happens in the in the torture example I think is that the um that uh your your what would normally be transparent becomes opaque and uh then becomes very conspicuous to you and and your uh way um uh dealing with the uh prediction errors that come with a pain experience and we've done some work on pain is um is by uh dissociating from it and so okay that's interesting so it's yes I mean people do talk about derealization for example as this kind of bubble wrap um that's there as a function to dissociate from Pain and from the anxiety of being eaten by a tiger or whatever in the savannah um so okay that's it's a kind of numbing of the pain the so you're arguing that the minus is still there it's just somewhat clouded by this objectification or uh opacity of what normally would be transparent so go back to melon's example of the two hands touching which he uses to illustrate how the body is both object and a subject of experience and ordinarily there's this what he calls a kaym so an intertwining of the subjects and the objects in the body um and what happens in um depersonalization derealization is then is that that this intertwining of subject and object now breaks down you have now a a kind of distancing of the subject from their body um and that's NE necessary in the torture example U

it's necessary to in some way disidentify with the pain experiences to treat them as some kind of objects that separate from you uh if you to be able to to deal with the pain uh so the derealization is a kind of adaptive response to what would otherwise be overwhelming where you can you can gain a distance from it and relate to it as as if it were not you that this was happening to as if it were an object that was separate from you this is this is this is very much Anna's account um and it feeds into this notion of kind of hyper Precision over um beliefs about the self model which leads to this kind of dissociation but again like the way that I was writing about it was not only that but also what um Dean and colleagues spoke about in 2020 which is they use this quote from a writer called Loft House 2014 saying the first time I remember feeling like the first time I can remember feeling like I didn't exist I was 15 I was sitting on a train and all of a sudden I felt like I'd been dropped into someone else's bodies my memories experiences and feelings uh and feelings the things that make up my intrinsic sense of meanness projected across my mind like fantasmagoria but I felt like they belong to someone else like I was experiencing life in the third person so this seems slightly different but I think the key thing to pull out from that quote is I felt like like these experiences belong to somebody else that I felt there is still the presence of mindness so it's me that this is happening to similarly in in examples of ego dissolution that happen yes that's what I was going to ask about as well psychedelics that George talks about in the losing ourselves paper and also in the another paper about ego dissolution I forget what it's called um there are a couple on there yeah yeah uh so you might think well ego dissolution is a kind of egoless experience this is how Raphael Miler describes uh ego dissolution in in extreme examples that there's just this complete uh loss of uh of ego of self um but uh I I think rather what happens in those examples is that uh there's you that's undergoing these these exp very strange experiences um you still present as the subject of them it's still U me that this is happening to I've taken the Psychedelic and now I'm undergoing this experience of uh loss of boundary between me and the world or this kind of feeling of Union and connectedness with nature um and that's an experience that I'm having so it's there's there's still that mindness that's part of the experience um but there's something that's lost as well which is this higher narrative uh layer of the self so instead of like identifying with who I have been in the past and then uh that ident identity um being brought into the framing of my experience I think George is right that there's this kind of loss of temporal depth so yeah influence of the narrative on your on your experience of your of who you are at that moment is now Stripped Away but still what's in place is the core or minimal self the mindness and for you that's just intrinsic to the experience

yeah it's part of the structure of the experience it's a phenomenological structure interesting described it called it IP of course Disturbed in in uh psychopathologies like schizophrenia or uh depersonalization derealization depression these are all disturbances of the self yes of that Ipsy which is part of our experiences our our perspective on the world so I guess my thinking here is that where Maybe I'm Wrong that where Dean and where I've divorced Dean George's and Anna's accounts there they can actually be tied under the single banner of uh distancing opacity hyper precision over self models and so on not that Dean not that George's account implies the total uh dissolving of the self model uh well you know there's there's some debate about this isn't there about whether the self model is entirely dissolved or not and George I think just talks about it in terms of this shrinking of the temporal depth uh this diminution of the temporal depth um so you you could ask well is there still something that is left of the self once that temporal depth is lost yes um I would tend to think about it in terms of different layers of the self model um and the the temporal the loss of temporal depth is describing the The Narrative layer of the self model um that could leave in place the bodily y core minimal self which is what is accounting for mindness and but it but it isn't in of itself according to you um sufficient for mindness in the sense that mindness is just part of the structure of experience so is what is there a distinction here between mindness and self modeling such that you could have dissolved entirely dissolved self models but you would still have experiences that by necessity are POS are owned by someone goes back to the same question about the relationship between phenomenology personal level explanation subpersonal processes basian mechanics and the way I think about that relationship is using the idea of generative passage from verella so mindness and phenomenal selfhood I don't think are um identical with um the generative model and the processes that are described using basian mechanics um they they those processes explain um in the sense of uh they give us a model that would allow us to bridge uh to phenomenology but but they don't reduce they don't you can't reduce the yes first person experience which is where we start from the lived experience to what can then be described modeled in third person terms using basian mechanics but if I were to say that I guess what I'm getting up to getting to here is you could claim there have been claims for example that interceptive inference underwrites the minimal self model such that if interceptive inference falls apart if there's if it's just completely unsuccessful such as in the torture chamber the minimal self model itself will dissolve if if you just grant me that uh causal relationship well the construct of the minimal self model is a a construct introduced by met singer yeah in order to um make sense of cognitive neuroscientific models of uh uh of

lived experience of phenomenal selfhood and of course met singer thinks that the we should be strong realists about the phenomenal self model as described in cognitive neuroscience and and we should say that that what we learn from cognitive Neuroscience is there's no self and that the that mindness is just a a construct of the brain yes it's hallucinated actually in the end a controlled hallucination um yes and you can but in the end you can lose it entirely yeah because it depends on and is identical with um inferential processes that are taking place in brain whereas my view is different from Thomas's in that I don't start from cognitive neuroscience and what can be modeled within cognitive Neuroscience I take as my starting point this is what verella meant when he said um that cognitive science should always uh go back to find a way back to lived experience because lived experience is where we start from and he's what we must always return to and this is from um his paper on um neurophenomenology so so the the idea there is that what's basic what we start from is lived experience um and then we have to try to find a way to explain that in the terms of cognitive neuroscience and that's what the phenomenal self model gives us and what basium mechanics give us they give us a way of explaining in naturalistic cognitive science terms what is described in phenomenology but what's real what's most real here what we start from is lived experience would you say that thas says that the phenomenal self model is identical to the inferences that the brain makes or is emergent over the inferential mechanics that are occurring in the brain as in one is causally related the other one's an identity depends what you mean by INF um emergent there so emergence can be understood in like a a weak sense of yeah um epistemically we we can't really predict from the the lower level the properties that we find at the higher level because of nonlinearity because of um uh know because there's there's something um nonadditive about the causes uh that means you get something qualitatively new once you move to the higher level yes and that can just be understood in a a purely epistemic way or it can be understood ontologically that there's something new that's brought into existence uh a self uh you get from U or a phenomenal self model which is just a construct of the brain but it's a higher level construct that emerges out of lower level processing as something ontologically new um and then you could say well what's the relationship between the phenomenal self model and the experience of being somebody or the experience of mindness and Thomas's answer to that last question is that the experience of being someone is an illusion it's a a the outcome of um the phenomenal self- modeling process which is in turn the outcome of inference and basian modeling he would now say I guess um so um emergence to answer your question can be understood in like a weak sense or it can be

understood in a stronger ontological sense and the stronger ontological notion doesn't decide in favor of the reality of the self because you could still believe that the phenomenal self model is strongly emergent even though in the end it's just an inferential construct of the brain and so whether you say emergence gets you a real self or just gets you uh an ersatz construct as Thomas argues I think doesn't fall out of the concept of emergence on its own okay that's that's really that's real food for thought um I'm going to have to want to differentiate out my position and the position of other people that are trying to do neurophenomenology from Thomas's position because we we think I'll speak for myself I think that the Sellarsian project of generative Transcendentalism means that what's real here is um being in the world or um lived experience the body subject and then the task for for anyone that is doing fore cognition that comes out of the inactive strand of fore cognition is to try to build a bridge back from lived experience to what can be described scientifically so thereby making room for lived experience Within science and that can then take a more elaborate form like John V's project of not only just trying to build room for lived experience within cognitive science but these rich Notions like wisdom and the sacred and yeah meaning in life and so on but yeah it's it's at least according to your the your own phenomenological Camp I'm right to say that the reduction to the science is a false move yeah I take fore cognition to be non-reductive and sometimes that that non-reduction move is is cashed out in terms of a strong notion of emergence so that connects to what I think you were trying to do so Evan Thompson will talk um neurophenomenology as committed to uh circular causality where at the the level of the higher levels that are to do with consciousness and lived experience loop back down and change the yes the lower level Dynamics so that's a a kind of downward causation or a strong form of emergence that he he he thinks you get from um from neurodynamics from self organization from autonomy which is where we started from um so neurophenomenology then would would get you a kind of metaphysics where uh there a strong form of emergence and Consciousness can actually make a difference to um neurodynamics to yeah as a higher level property it can loop back down and and change the faster changing processes in the brain yes well this this has its resonance to use a Rel relevant term with active inference um but yeah as you say it does there's this idea of enslaving and synergetics and that Carl has been influenced by it comes from Herman Haken where uh there's there's like ensemble dynamics that unfold over slower time scales that loop back and change the faster time scales so causation that's been described there just get away from the analytic philosophy just more colloquially would you ever use the term self-modeling for you does it make does it does it make sense in

terms of the uh phenomenological starting point to even talk about self models or is it does it does it make more sense to talk about intentionality and mindness and so on so I again we have this necessity to to keep apart the phenomenology which we're trying to bridge to where I think Notions like mindness and intentionality are have their home yeah and the the the modeling that we're doing when we use active inference models to explain uh there I don't where we also talk about models and um we talk about a generative model and and we then give intentional semantic glosses on that generative model but that last move I I resist making and I say know that intentionality always belongs within experience which we can then explain using basian mechanics which is non-intentional non-representational I me I meant more maybe I should have said more this is like metacognition because metacognition in a way is it feels like a form of intentional self-modeling or phenomological self- modeling in the sense of in lived experience you are reflecting on yourself yeah is that not a self model which is grounded in phenomenological intentionality so using uh Thomas's framework we can talk about the phenomenal self model and we can say within that there's a minimal self and then there's the four epistemic agent model yeah yeah the epistemic agent model so these these are all ways of talking about the self model within uh active inference um and I don't of course I don't deny that we can talk about the self model there in fact I in the paper you talked about I do something that no one else has picked up on and no one else seems to like which is to say that all generative models are actually self models because the what's being modeled by a generative model is an agent uh so going back to the uh the agent is a model of its environment and that Theorem and so on and and the agent in in active uh cognitive science in active philosophy is uh a point of view a perspective on the world relative to which you can make sense of the world and make distinctions between what's good for me what's bad for me and so uh I want to argue that um right at the core of all generative modeling because the generative model at least when it's applied to life is the model of an agent a minimal agent is a self- model uh so generative models are always self models I want to argue it's not that there's some part of a a generative model at the higher layers of it where we can start to say well that's the self model or in interceptive inference say and say that's the self model I think um mik Allen and and Manos Securus have a beautiful figure actually where they have like the self model right at the core of uh of active inference and that's to me is right that they talk about the first prior yeah the body is the first prior Yeah Anna Cher and and um Catalina also talk about the the and Manos also talk about the self as a first prior but they understand the self model here in a more relational ways into subjective so thinking about attachment Dynamics

for instance but all of that work for me indicates that you can't really take the idea of the agent as a model of its environment without thinking that the model there is a self model so I'm very serious I think the self model construct is extremely important in thinking about how the agent could be a model of its environment but the move that Thomas makes then of just reducing the self as a phenomenological experience to the self modeling I want to resist because I don't like his reductionism and I certainly don't like his move of saying the self doesn't exist self is not real and I don't think Buddhism really uh substantiates that idea and I don't think uh cognitive science supports it either so I would resist a kind of move to a no self Theory okay I'm going to call I'm going to say that that's that's good for me because uh We've covered so much ground and we could go on for hours more but tired you're tired so well I just there's so much to think about um you know in my to be candid I mean I'm coming out of the metzger school which is informed Act of inference so it's difficult at a personal level to real time shift one's Paradigm because maybe some of the Bear in mind that there's U there are other ways of using active inference here that don't necessarily leads to a no self Theory and well yeah I mean I I I I wouldn't is there a difference between the no self and the constructed self uh yes and that you can think that constructions are real U whereas Thomas want to say no if the self is constructed inferentially yeah then we should infer from that that that no one ever was or had a self yes I've seen him say that yeah so I agree that the self is a construct so Evan Thompson what Evan says yeah yeah he has this very nice idea in in his work on uh on Eastern philosophy that he's why I am not a Buddhist yeah where he thinks that the self is a construct but the layers of the self that construct the self come from things like the body from our into subjective relations like Katarina and Manos aru from our uh being uh members of a culture where other people can understand us in a particular way so then we can take on board how we're perceived or understood by others so these are all layers that lead to the construction of the self um but we shouldn't then infer that the self isn't real just because it's constructed or no one ever was or had a self that just doesn't follow and I I don't want to like go on the side of instrumentalist here and say if it's a modeling tool then what's being modeled is not anything real right so Michael and I have done work on this with Ian Robertson where we say that that amounts to making what we call the the literalist fallacy so the fallacy of thinking that in order for a model to be a model of Something Real the model has to be taken literally as describing what is literally happening in the world and we think that's not true you can be a realist about these models without um without thinking that what's being described is something that's literally real and so sure models involve you making all kinds of imaginative

leaps from what your modeling to the Target phenomena yeah that that that's really useful I you know I feel like I've been slightly converging on this in my own thinking that there's a something that I've been coming to find is that there's a difference this might be not entirely the same but it's adjacent there's a difference between maybe what the basian mechanics permits and I use that word knowing that it's probably the wrong word but permits in terms of the phenomenology and there's an actual but there's a difference between that and the actual phenomenology itself and I think that might be akin to yeah that's exactly what I'm saying so there's the model that we use as scientists and then there's the target system that we're modeling which is the system as just well phenomenologists wouldn't call it a system but Liv the experience as it's described which is what we're trying to do and we're trying to build this generative passage from the modeling to phenomenology um and those are different one is the model the other is the target of the model yeah yeah yeah yeah the Target phology or it can be the brain because these are both targets that we're trying to use the generative modeling to capture but the relationship here is between the model and the target system oh that's so useful that's so useful again it's not helped by the fact that it's called a phenomenal self model is it I mean uh that's a way of really deflating that Gap but of course for metzinger he does that for a good reason because he thinks no one ever was or had a self so all that is to the self is the phenomenal self model whereas I'm saying no phenomenologically the self belongs in phenomenology and it's real because that's where it comes from but isn't it a bit odd to call it a phenomenal self model and then say there's nothing there's no self given that it's phenomenal does that not strike you as a slight contradiction because we take phenomenal the phenomenal reality to be real in of itself regardless of how we end up demarcating it and reifying it we take lived experiences the stuff idea of the ego Tunnel right so yeah and it's a bit like the pl Plato's Cave myth so yes the self appears to us and it appears to be real but the argument is then it's just an appearance it's just a and where does that appearance come from it comes from the construction of this phenomenal self model that's enough for me that still seems to be like what you're talking about in terms of mindness subjectivity intentionality which to me seem to be the things that matter in the end yeah interesting I think comes back to is whether you're a physicalist or not so if you if you think that reality fundamentally is just what can be described and explained by physics which I think Thomas would in the end endorse yes uh the phenomenal self model is going to be reducible to um to neurobiological processes and those are going to In the End come back to something physical whereas phenomenologists are not physicalists they're non-reductive um yeah yeah yeah yeah naturalists yeah yeah they try to

have a softer form of naturalism John fi talks about Transcendent naturalism in some of his work that's a that's because he has this he has in view uh trying to get to the wisdom traditions in the end from these ideas but um but these softer forms of naturalism I think are alive possibilities for us as well when we're using active inference and the the free energy principle we don't have to be physicalists whereas Alex yakob Hoy you know they're they're long time on uh card holding physicalists um and the free energy principle goes nicely with physicalism because there's so many of these ideas are coming from physics and um so you might think well it follows from where all of these ideas are being drawn from that we can we should end up being physicalists about uh anything that's described in phenomenology which means being reductionists but I just don't think that that's right and so so going forward how how would well not going forward in your present work and past work and forward work how would you talk what words would one use to talk about bridging this Gap because we want to keep the basing mechanics and we want to keep the phenomenology and we want to find an explanation that's non-reductive so what is the what what you know this is taking me back to philosophy when I was at school what are the relations that we're trying to describe here are these causal relations are these like are these you know they're not epip phenomenal but what so how how what's the best way of talking about them from your perspective uh well uh I see myself as doing neurop phenomenology so what that means is that um I think there's a necessity when we're explaining phenomena that ATT tied to a lived experience like Consciousness the self and disturbances that can happen in mental illness uh of making reference to first second person experience and and that then means using um methods for studying first person experience like uh phenomenology but also uh contemplative methods like meditation mindfulness because they involve something that is akin to uh the phenomenological reduction as it was practiced by phenomenology so what we then need to try to do is to find ways of bridging between what can be described in first person terms and what can be described third person what's the relationship between those well a reductionist would say that the first person reduces to the third person whereas uh I try to uh have a metaphysics where uh it's non-reductive so what does that look like well I don't think there's anything outside of science that we're dealing with here so there nothing like metaphysically spooky being introduced here I wouldn't want to say but we need a more relaxed naturalism um what is the metaphysics of that going to involve well it involves having a richer notion of causality for instance than just efficient causality because we're thinking about self-organizing systems now so we need to have what uh Al Alicia wero in her book um context matters for

everything or something right constraints yeah enabling constraints yeah yeah so we need that we need a notion of enabling constraints for instance here um we need self-organization we need to have a kind of teleology here so in discussions of the free energy principle people sometimes wonder about um what's the reality of uh of the what Spinoza called the *canus* is part of what we're looking at when we try to model agents as free energy minimizing systems and axol said no it's not because um the existence of the agent is a consequence of free energy minimization it's not you don't get an explanation of why agents exist out of free energy minimization and I want to say something different from Axel there that um that living systems do indeed have this drive to maintain themselves this *canatas* so the metaphysics that we need here is something that looks a lot like Spinoza's metaphysics that you've got a you've got a substance um I I don't really like substance talks or that part of ity and resist and go to processes but sure but you do need this idea of a *canus* I think um in accounting for what makes the difference between a pebble and a living system say and we can make sense of in terms of free energy minimization as well so the metaphysics that comes out of this is not physicalist metaphysics it's something that has room for there being a difference between living systems and Pebbles um and that we can make sense of Beyond just a physics in terms of Basi and mechanics so look at the paper about strange particles for instance yeah so um and look at the work that Michael and I did about the difference between mere active inference and adaptive active inference and and there's there's an account there a princip account that comes out of active inference about the difference between a mere physical system and a living system a biological agent this goes Loops back to where we started with a yeah of course of course I'm just so aware that Carl would just say well this is this is just and I'm using his hand signals uh you know this is the depth of temporal planning that's again what you're saying difference yeah but you're saying that's that's the basian mechanics but you that's not everything there's a non-reductive phenomenology on top of it yeah so I do want to say that that uh that it isn't just about the the temporal depth of the generative model that's difference between the a being that has a *canus* a drive to maintain itself and and uh yeah because the model here is just a model of that kind of thing that kind of being yeah yeah yeah yeah oh Julian gosh I've learned so much I thought I came in knowing roughly what I was talking about and I've left ever so confused and in doubt of myself but nonetheless it's been wonderful and or inspiring and Incredibly informative should do to you they should just leave you in a a bit of a confused state where you then try sort through your confusions and make sense of them and no no not at all it's been incredibly useful I feel like I've kind of um cheated my way

six months ahead of my education in in three hours so that's no bad thing it's it's no just super informative um what can people look forward to in terms of your upcoming work yeah so something we didn't get to talk about at all which is a shame is the work we've been doing on play and playfulness yes I know yes yeah a Dynamics aerodynamics came up very briefly so I've been working with Mark Miller and in uh arus University with uh Mark Anderson there um we've been developing ideas around uh why it's fun and and pleasurable to learn through play activities um so together with Mark Anderson we have a paper that's uh now under revision with topics in cognitive science and in a special issue about developments of development of models of the world um and we talk about the contribution of play to uh in child development in that paper using the our ideas about aerodynamics and carrying on our account of why it's fun to learn through uh play yes so that's one of the things that we're working on um and then together with my colleague Daman Denise here in Amsterdam we're starting a new center on uh uh this interface between Psychiatry Psychotherapy and philosophy fantastic my work as part of that will be using active inference to try to uh understand Psychotherapy say in the context of psychedelic Psychotherapy for instance so I'm doing work on psychedelics that come out of that but also trying to apply active inference to to make make sense of um mental illness more generally and finally know how things can go well for people so that's the work that I've been doing with Mark Miller on uh well-being and meaning in life um that's going to continue to be a strand of research to uh to try to make sense of how uh how people can be happy and be well using these tools of uh of active inference wonderful well I'm gonna need to go take a nap because I've had my worldview just demolished and I didn't even know that it was problematic but it's fine um I'm sure we'll stay in touch um I'd love to you're it's true inspiration and the the wealth of knowledge is just unbelievable so thank you again for giving me three hours which has felt like 15 minutes my final question is a tongue in cheek one but you've got it I'm giving you three phenomenologists and you have to choose one uh hle Merlo Ponte wow uh well I'm going to go for Merlo Ponte because he's like a a combination of H and haiger both of them yeah okay and he's French so back your your your Franco File comment earlier on that everything from France is wonderful including French philosophy so apart when it Reigns in June according to my model who's French rather than german yes yeah absolutely uh and a bit more legible of course but the style is just so much more pleasurable to engage with of his writing than heidig has his own style but it's uh it's much harder and H is he's a mathematician so there's few hyphens in Merlo Ponte despite the hyphen in the name the H yes Julian thank you so much um I think we

probably there's been a couple of WiFi on and off but this is the beauty of editing as I will go in and scrub it all out but um great genuinely on my part it's been an absolute pleasure um I know Daniel on the rest of the inference rest clear enough you're able to follow it all make a story out of it I hope so uh but yes for me and the rest of the team here at the Institute thank you so much my pleasure